

Chapter 25

Statistics

Solutions

SECTION - A

Objective Type Questions (One option is correct)

1. The mean deviation of the numbers 11, 12, 13, 14, 15 about the mean is

(1) 1 (2) 1.2 (3) 1.5 (4) 2

Sol. Answer (2)

$$\bar{x} = \frac{11+12+13+14+15}{5} = \frac{65}{5} = 13$$

x_i	$d_i = x_i - \bar{x} = x_i - 13 $
11	2
12	1
13	0
14	1
15	2
Total	$\sum d_i = 6$

$$\sum d_i = 6 \text{ and } n = 5$$

$$\therefore \text{M.D.} = \frac{1}{n} \sum |d_i| = \frac{6}{5} = 1.2$$

2. The mean deviation from the median is

(1) Equal to that measured from any another value (2) Maximum if all observations are positive
(3) Greater than that measured from any other value (4) Less than that measured from any other value

Sol. Answer (4)

3. The mean deviation of the data 5, 7, 4, 9, 3, 14 from the mean is

(1) 2 (2) 3 (3) 4 (4) 7

Sol. Answer (2)

$$\bar{x} = \frac{3+4+5+7+9+14}{6} = \frac{42}{6} = 7$$

x_i	$d_i = x_i - \bar{x} = x_i - 7 $
3	4
4	3
5	2
7	0
9	2
14	7
Total	$\sum d_i = 18$

$$\sum d_i = 18, \quad n = 6$$

$$\text{M.D.} = \frac{1}{n} \sum |d_i| = \frac{18}{6} = 3$$

4. The mean deviation of the data 7, 4, 9, 12, 14, 20 from the mean is

(1) 2.7

(2) 3.7

(3) 4.3

(4) 5.1

Sol. Answer (3)

$$\bar{x} = \frac{4+7+9+12+14+20}{6} = \frac{66}{6} = 11$$

x_i	$d_i = x_i - \bar{x} = x_i - 11 $
4	7
7	4
9	2
12	1
14	3
20	9
Total	26

$$\sum d_i = 26 \text{ and } n = 6$$

$$\text{M.D.} (\bar{x}) = \frac{1}{n} \sum |d_i| = \frac{26}{6} = 4.3$$

5. Mean deviation for n observations $x_1, x_2, \dots, x_n$ from their mean $\bar{x}$ is given by

(1) $\sum_{i=1}^n (x_i - \bar{x})$

(2) $\frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$

(3) $\sum_{i=1}^n (x_i - \bar{x})^2$

(4) $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$

Sol. Answer (2)

6. When tested, the lines (in hours) of 5 bulbs were noted as follows:

1357, 1090, 1666, 1494, 1623

The mean deviations (in hours) from their mean is

- (1) 178 (2) 179 (3) 220 (4) 356

Sol. Answer (1)

$$\bar{x} = \frac{1357 + 1090 + 1666 + 1494 + 1623}{5} = \frac{7230}{5} = 1446$$

x_i	$ x_i - \bar{x} $
1357	89
1090	356
1666	220
1494	48
1623	177
Total	890

$$\text{M.D. } (\bar{x}) = \frac{\sum |d_i|}{N} = \frac{890}{5} = 178$$

7. The mean deviation about the median for the following data is:

7, 5, 4, 3, 4, 3, 2, 11, 19, 21, 15

- (1) 5.04 (2) 5.18 (3) 6.02 (4) 4.32

Sol. Answer (2)

2, 3, 3, 4, 4, 5, 7, 11, 15, 19, 21

$$\text{Median} = \left(\frac{11+1}{2} \right)^{\text{th}} \text{ observation} = 6^{\text{th}} \text{ observation} = 5$$

The absolute values of respective deviations from the median;

3, 2, 2, 1, 1, 0, 2, 6, 10, 14, 16

$$\therefore \sum_{i=1}^{11} |x_i - M| = 57$$

$$\text{M.D. } (M) = \frac{1}{11} \sum_{i=1}^{11} |x_i - M| = \frac{57}{11} = 5.18$$

8. The mean deviation about the median for the following data is:

210, 208, 388, 238, 510, 359, 412

- (1) 95 (2) 93.4 (3) 79.8 (4) 56.7

Sol. Answer (2)

208, 210, 238, 359, 388, 412, 510

$$\text{Median} = \left(\frac{7+1}{2} \right)^{\text{th}} \text{ observation} = 4^{\text{th}} \text{ observation} = 359$$

The absolute values of respective deviations from the median,

151, 149, 121, 0, 29, 53, 151

$$\therefore \sum_{i=1}^7 |x_i - M| = 654$$

$$\text{M.D. (M)} = \frac{1}{7} \sum_{i=1}^7 (x_i - M) = \frac{654}{7} = 93.42$$

9. Following are the marks obtained by 9 students in a mathematics test 50, 69, 20, 33, 53, 39, 40, 65, 59.

The mean deviation from the median is

- (1) 9 (2) 10.5 (3) 12.67 (4) 14.76

Sol. Answer (3)

20, 33, 39, 40, 50, 53, 59, 65, 69

$$\text{Median} = \left(\frac{9+1}{2} \right)^{\text{th}} \text{ observation} = 5^{\text{th}} \text{ observation} = 50$$

The absolute values of respective deviations from the median:

$\therefore$ 30, 17, 11, 10, 0, 3, 9, 15, 19

$$\therefore \sum_{i=1}^9 |x_i - M| = 114$$

$$\text{M.D. (M)} = \frac{1}{9} \sum_{i=1}^9 |x_i - M| = \frac{1}{9} \times 114 = 12.67$$

10. The mean deviation from the median of the following data is:

49, 30, 36, 32, 12, 52, 13, 23, 42, 25

- (1) 11.4 (2) 10.8 (3) 12.6 (4) 13.5

Sol. Answer (2)

12, 13, 23, 25, 30, 32, 36, 42, 49, 52

$$\text{Median} = \frac{\left(\frac{10}{2} \right)^{\text{th}} \text{ observation} + \left(\frac{10}{2} + 1 \right)^{\text{th}} \text{ observation}}{2} = \frac{30 + 32}{2} = 31$$

The absolute values of respective deviations from the median:

19, 18, 8, 6, 1, 1, 5, 11, 18, 21

$$\therefore \sum_{i=1}^{10} |x_i - M| = 108$$

$$\text{M.D. (M)} = \frac{1}{N} \sum_{i=1}^{10} |x_i - M| = \frac{108}{10} = 10.8$$

11. The mean deviation about the mean for the following data is:

x_i	20	22	25	27
f_i	3	8	6	3

(1) 4.283

(2) 3.463

(3) 5.762

(4) 2.075

Sol. Answer (4)

x_i	f_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
20	3	60	3.25	9.75
22	8	176	1.25	10
25	6	150	1.75	10.5
27	3	81	3.75	11.25
Total	20	467		41.5

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{467}{20} = 23.25$$

$$\text{M.D. } (\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} = \frac{41.5}{20} = 2.075$$

12. The mean deviation about the mean for the following data is:

x_i	5	8	9	10	12
f_i	1	3	1	3	2

(1) 2.4

(2) 1.6

(3) 0.8

(4) 3.2

Sol. Answer (2)

x_i	f_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
5	1	5	4.2	4.2
8	3	24	1.2	3.6
9	1	9	0.2	0.2
10	3	30	0.8	2.4
12	2	24	2.8	5.6
	10	92		16

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{92}{10} = 9.2$$

$$\text{M.D. } (\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} = \frac{16}{10} = 1.6$$

13. The mean deviation about the mean for the following data is:

x_i	4	6	7	9	12
f_i	3	1	2	4	5

(1) 3.08

(2) 8.53

(3) 2.56

(4) 3.75

Sol. Answer (3)

x_i	f_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
4	3	12	4.53	13.59
6	1	6	2.53	2.53
7	2	14	1.53	3.06
9	4	36	0.47	1.88
12	5	60	3.47	17.35
	15	128		38.41

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum f_i} = \frac{128}{15} = 8.53$$

$$\begin{aligned} \text{M.D. } (\bar{x}) &= \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} \\ &= \frac{38.41}{15} = 2.56 \end{aligned}$$

14. The mean deviation about the median for the following data:

x_i	3	4	5	6	7	8	9
f_i	4	3	2	1	4	5	1

(1) 2.6

(2) 1.85

(3) 3.26

(4) 2.31

Sol. Answer (2)

x_i	f_i	c.f.	$ x_i - M $	$f_i x_i - M $
3	4	4	3.5	14
4	3	7	2.5	7.5
5	2	9	1.5	3
6	1	10	0.5	0.5
7	4	14	0.5	2
8	5	19	1.5	7.5
9	1	20	2.5	2.5
	$N = 20$			37

$$\text{Median} = \frac{\left(\frac{20}{2}\right)^{\text{th}} + \left(\frac{20}{2} + 1\right)^{\text{th}}}{2} = \frac{6+7}{2} = 6.5$$

$$\text{M.D. (M)} = \frac{1}{N} \sum_{i=1}^7 f_i |x_i - M| = \frac{37}{20} = 1.85$$

15. The mean deviation about the median for the following data:

x_i	4	8	12	16	20	24
f_i	3	2	7	5	2	1

(1) 4.2

(2) 4

(3) 1.5

(4) 5.8

Sol. Answer (2)

x_i	f_i	c.f.	$ x_i - M $	$f_i x_i - M $
4	3	3	8	24
8	2	5	4	8
12	7	12	0	0
16	5	17	4	20
20	2	19	8	16
24	1	20	12	12
	$N = 20$			80

$$\begin{aligned} \text{Median} &= \frac{\left(\frac{20}{2}\right)^{\text{th}} + \left(\frac{20}{2} + 1\right)^{\text{th}} \text{ observation}}{2} \\ &= \frac{12 + 12}{2} = 12 \end{aligned}$$

$$\begin{aligned} \text{M.D. (M)} &= \frac{1}{N} \sum_{i=1}^n f_i (x_i - M) \\ &= \frac{80}{20} = 4 \end{aligned}$$

16. The mean deviation about the mean for the following data:

Marks obtained	10-20	20-30	30-40	40-50	50-60
Number of students	8	6	12	17	7

(1) 12.06

(2) 10.67

(3) 16.05

(4) 35.2

Sol. Answer (2)

Marks obtained	Number of students f_i	Mid-points x_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
10 – 20	8	15	120	21.8	174.4
20 – 30	6	25	150	11.8	70.8
30 – 40	12	35	420	1.8	21.6
40 – 50	17	45	765	8.2	139.4
50 – 60	7	55	385	18.2	127.4
Total	50		1840		533.6

$$N = \sum_{i=1}^5 f_i = 50$$

$$\bar{x} = \frac{\sum_{i=1}^5 f_i x_i}{N} = \frac{1840}{50} = 36.8$$

$$\text{M.D. } (\bar{x}) = \frac{1}{N} \sum_{i=1}^5 f_i |x_i - \bar{x}| = \frac{533.6}{50} = 10.67$$

17. The mean deviation about the mean for the following data is:

Class	30–40	40–50	50–60	60–70	70–80	80–90
Frequency	2	8	7	5	8	10

(1) 10.85

(2) 14.28

(3) 13.15

(4) 14.5

Sol. Answer (2)

Class	Frequency f_i	Mid-points x_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
30 – 40	2	35	70	29.75	59.5
40 – 50	8	45	360	19.75	158
50 – 60	7	55	385	9.75	68.25
60 – 70	5	65	325	0.25	1.25
70 – 80	8	75	600	10.25	82
80 – 90	10	85	850	20.25	202.5
Total	40		2590		571.5

$$\bar{x} = \frac{2590}{40} = 64.75$$

$$\text{M.D. } (\bar{x}) = \frac{571.5}{40} = 14.28$$

18. The mean deviation about the mean for the following data :

Class	100–120	120–140	140–160	160–180	180–200
Frequency	2	1	3	2	2

(1) 25

(2) 22.4

(3) 31.5

(4) 41

Sol. Answer (2)

Class	Frequency f_i	Mid-points x_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
100 – 120	2	110	220	42	84
120 – 140	1	130	130	22	22
140 – 160	3	150	450	2	6
160 – 180	2	170	340	18	36
180 – 200	2	190	380	38	76
Total	10		1520		224

$$\bar{x} = \frac{1520}{10}$$

$$= 152$$

$$\text{M.D. } (\bar{x}) = \frac{224}{10}$$

$$= 22.4$$

19. The mean deviation about the median for the following data is:

Class	10–20	20–30	30–40	40–50	50–60
Frequency	2	4	6	8	10

(1) 12.5

(2) 13.8

(3) 10.25

(4) 14.25

Sol. Answer (3)

Class	Frequency f_i	c.f.	Mid-points x_i	$ x_i - M $	$f_i x_i - M $
10 – 20	2	2	15	28.75	57.5
20 – 30	4	6	25	18.75	75
30 – 40	6	12	35	8.75	52.5
40 – 50	8	20	45	1.25	10
50 – 60	10	30	55	11.25	112.5
Total	$N = 30$				307.5

The class interval containing $\frac{N}{2}$ item i.e., 15th item = 40 – 50

$$\begin{aligned}
 \text{Median} &= l + \frac{\frac{N}{2} - c}{f} \times h \\
 &= 40 + \frac{15 - 12}{8} \times 10 \\
 &= 40 + 3.75 \\
 &= 43.75
 \end{aligned}$$

$$\text{M.D. (M)} = \frac{307.5}{30} = 10.25$$

20. The mean deviation about median for the following data:

Marks	10-20	20-30	30-40	40-50	50-60
Number of Students	5	4	9	6	6

(1) 12

(2) 11

(3) 13

(4) 18

Sol. Answer (2)

Marks	Number of Students f_i	Cumulative frequency c.f.	Mid-points (x_i)	$ x_i - M $	$f_i x_i - M $
10 - 20	5	5	15	21.67	108.35
20 - 30	4	9	25	11.67	46.68
30 - 40	9	18	35	1.67	15.03
40 - 50	6	24	45	8.33	49.98
50 - 60	6	30	55	18.33	109.98
Total	$N = 30$				330.02

The class interval containing $\frac{N^{\text{th}}}{2}$ item i.e., 15th item = 30 - 40

$$\begin{aligned}
 \text{Median} &= l + \frac{\frac{N}{2} - c}{f} \times h \\
 &= 30 + \frac{15 - 9}{9} \times 10 \\
 &= 30 + 6.67 \\
 &= 36.67
 \end{aligned}$$

$$\text{M.D. (M)} = \frac{330.02}{30} = 11.00 = 11$$

21. The mean deviation about the median for the following data is:

Class	0-10	10-20	20-30	30-40	40-50
Frequency	7	16	13	21	13

(1) 11.12

(2) 15.06

(3) 19.6

(4) 13.52

Sol. Answer (1)

Class	Frequency (f_i)	Cumulative frequency	Mid-points (x_i)	$ x_i - M $	$f_i x_i - M $
0 – 10	7	7	5	24.231	169.617
10 – 20	16	23	15	14.231	227.696
20 – 30	13	36	25	4.231	55.003
30 – 40	21	57	35	5.769	121.149
40 – 50	13	70	45	15.769	204.997
Total	$N = 70$				778.462

The class interval containing $\frac{N}{2}$ term = 20 – 30

$$\begin{aligned} \text{Median} &= l + \frac{\frac{N}{2} - c}{f} \times h \\ &= 20 + \frac{35 - 23}{13} \times 10 \\ &= 29.231 \end{aligned}$$

$$\text{M.D. (M)} = \frac{778.462}{70} = 11.12$$

22. The mean of 7 observations is 11. If six of the observations are 2, 6, 8, 10, 12, 18 then remaining one observation is

(1) 18 (2) 21 (3) 14 (4) 16

Sol. Answer (2)

$$\bar{x} = \frac{2 + 6 + 8 + 10 + 12 + 18 + x}{7}$$

$$\Rightarrow 77 - 56 = x$$

$$\therefore x = 21$$

23. Median of the data 10, 21, 5, 6, 4, 12, 14, 3, is

(1) 16 (2) 8 (3) 4 (4) None of these

Sol. Answer (2)

24. Mean deviation about median for the following data 2, 5, 7, 2, 4, 6, 10, 12, 5

(1) 2.4 (2) 3 (3) 1.9 (4) None of these

Sol. Answer (1)

25. The mean of 5 observations 1, 2, 6, x and y is 4.4, then x + y is

(1) 4 (2) 9 (3) 15 (4) 13

Sol. Answer (4)

26. If $\sum_{i=1}^{10} x_i = 60$ and $\sum_{i=1}^{10} x_i^2 = 360$ then $\sum_{i=1}^{10} x_i^3$ is

- (1) 2160 (2) 3250 (3) 3360 (4) None of these

Sol. Answer (1)

27. The variance of the following data is

6, 7, 8, 12, 13, 15

- (1) 12.05 (2) 11.14 (3) 14.08 (4) 12.15

Sol. Answer (2)

$$\bar{x} = \frac{6+7+8+12+13+15}{6} = \frac{61}{6} = 10.1$$

x_i	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$
6	4.1	16.81
7	3.1	9.61
8	2.1	4.41
12	1.9	3.61
13	2.9	8.41
15	4.9	24.01
Total		66.86

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^6 (x_i - \bar{x})^2 = \frac{66.86}{6} = 11.14$$

28. If the variance of a data is 361, then the standard deviation of the data is

- (1) 16 (2) 17 (3) 18 (4) 19

Sol. Answer (4)

$$\text{Standard deviation } \sigma = \sqrt{\text{variance}} = \sqrt{361} = 19$$

29. Let a, b, c, d, e be the observations with mean m and standard deviation s . The standard deviation of the observations $a + k, b + k, c + k, d + k, e + k$ is ($k > 0$)

- (1) $m + k$ (2) $s + k$ (3) s (4) ks

Sol. Answer (3)

Adding a positive number to each observation, does not effects standard deviation.

30. The standard deviation of the data 6, 5, 9, 13, 12, 8, 10 is

- (1) $\sqrt{\frac{52}{7}}$ (2) $\sqrt{6}$ (3) $\frac{52}{7}$ (4) 6

Sol. Answer (1)

$$\bar{x} = \frac{6+5+9+13+12+8+10}{7} = \frac{63}{7} = 9$$

x_i	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$
6	-3	9
5	-4	16
9	0	0
13	4	16
12	3	9
8	-1	1
10	1	1
Total		52

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{52}{7}$$

$$\sigma = \sqrt{\frac{52}{7}}$$

31. The standard deviation of the data 12, 14, 18, 19, 21, 36 is

(1) 7.76

(2) 8.02

(3) 8.5

(4) 7.2

Sol. Answer (1)

$$\bar{x} = \frac{120}{6} = 20$$

The square of deviations about mean are 64, 36, 4, 1, 1, 256

$$\sigma^2 = \frac{362}{6} = 60.33 \Rightarrow \sigma = 7.76$$

32. Let $x_1, x_2, \dots, x_n$ be n observations and $\bar{x}$ be their arithmetic mean. The formula for the standard deviation is given by

(1) $\sum (x_i - \bar{x})^2$

(2) $\frac{\sum (x_i - \bar{x})^2}{n}$

(3) $\sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$

(4) $\sqrt{\frac{\sum x_i^2}{n} + \bar{x}^2}$

Sol. Answer (3)

33. The mean of 100 observations is 50 and their standard deviation is 5. The sum of all squares of all the observations is

(1) 500000

(2) 250000

(3) 252500

(4) 255000

Sol. Answer (3)

$$\sigma = \sqrt{\frac{\sum_{i=1}^{100} x_i^2}{100} - \bar{x}^2}$$

$$\Rightarrow 25 = \frac{\sum_{i=1}^{100} x_i^2}{100} - 2500$$

$$\Rightarrow \frac{\sum_{i=1}^{100} x_i^2}{100} = 2525$$

$$\therefore \sum_{i=1}^{100} x_i^2 = 252500$$

34. The mean of 50 observations is 25 and their standard deviation is 4, then the sum of squares of all the observations is

(1) 32050 (2) 33250 (3) 320500 (4) 753200

Sol. Answer (1)

$$\text{Variance} = \frac{\sum_{i=1}^{50} (x_i)^2}{n} - (\bar{x})^2$$

$$\Rightarrow 16 = \frac{\sum_{i=1}^{50} (x_i)^2}{50} - (25)^2$$

$$\Rightarrow \sum_{i=1}^{50} (x_i)^2 = 641 \times 50$$

$$\therefore \sum_{i=1}^{50} (x_i)^2 = 32050$$

35. Let x_1, x_2, x_3, x_4, x_5 be the observations with mean m and standard deviation s . The standard deviation of the observations $kx_1, kx_2, kx_3, kx_4, kx_5$ is

(1) $k + s$ (2) $\frac{s}{k}$ (3) ks (4) s

Sol. Answer (3)

If k is multiplied to each observation then standard deviation becomes k times.

36. Standard deviation for first 10 natural numbers is

(1) 2.87 (2) 3.25 (3) 6.35 (4) 5.5

Sol. Answer (1)

$$\bar{x} = \frac{(n+1)}{2} = \frac{11}{2} = 5.5$$

The deviation about mean are

$-4.5, -3.5, -2.5, -1.5, -0.5, 0.5, 1.5, 2.5, 3.5, 4.5$

$$\text{So, } \sigma^2 = \frac{82.5}{10} = 8.25$$

$$\sigma = 2.87$$

37. Consider the first 10 positive integers. If we multiply each number by -1 and then add 1 to each number, the variance of the numbers so obtained is

(1) 8.25 (2) 3.62 (3) 2.25 (4) 3.15

Sol. Answer (1)

x_i	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$
0	4.5	20.25
-1	3.5	12.25
-2	2.5	6.25
-3	1.5	2.25
-4	0.5	0.25
-5	-0.5	0.25
-6	-1.5	2.25
-7	-2.5	6.25
-8	-3.5	12.25
-9	-4.5	20.25
		82.5

$$\bar{x} = -4.5$$

$$\sigma^2 = \frac{\sum_{i=1}^{10} (x_i - \bar{x})^2}{10} = \frac{82.5}{10} = 8.25$$

38. The following relates to a sample of size 60:

$$\sum x^2 = 18000, \sum x = 960. \text{ The variance is}$$

(1) 6.63 (2) 16 (3) 22 (4) 44

Sol. Answer (4)

$$\begin{aligned} \sigma^2 &= \frac{\sum x^2}{n} - \bar{x}^2 \\ &= \frac{18000}{60} - \left(\frac{960}{60}\right)^2 \\ &= 300 - 256 \\ &= 44 \end{aligned}$$

39. The following information relates to a sample of size 100:

$$\sum x^2 = 20000 \text{ and } \sum x = 1260. \text{ The variance is}$$

(1) 55 (2) 21 (3) 41.24 (4) 38.65

Sol. Answer (3)

$$\begin{aligned}\sigma^2 &= \frac{\sum x^2}{n} - \bar{x}^2 \\ &= \frac{20000}{100} - \left(\frac{1260}{100}\right)^2 \\ &= 200 - 158.76 \\ &= 41.24\end{aligned}$$

40. Standard deviation of the data 2, 4, 5, 6, 8, 17 is 23.33. Then standard deviation of 4, 8, 10, 12, 16, 34 will be

- (1) 23.23 (2) 25.23 (3) 46.66 (4) 48.66

Sol. Answer (3)

When each observation is multiplied by 2, then standard deviation is also multiplied by 2.

41. The variance of 20 observations is 5. If each observation is multiplied by 2, then the variance of the resulting observations is

- (1) 10 (2) 20 (3) 25 (4) 12.5

Sol. Answer (2)

When each observation is multiplied by 2, then variance becomes 4 times.

42. If the standard deviation of the numbers 2, 3, $2x$ and 11 is 3.5, the possible value of x is

- (1) 2 (2) 5 (3) 3 (4) 4

Sol. Answer (3)

$$\begin{aligned}\sigma^2 &= \frac{\sum x^2}{n} - (\bar{x})^2 \\ (3.5)^2 &= \frac{(4 + 9 + 4x^2 + 121)}{4} - \left(\frac{8 + x}{2}\right)^2 \\ \Rightarrow \frac{49}{4} &= \frac{134 + 4x^2}{4} - \frac{64 + x^2 + 16x}{4} \\ \Rightarrow 3x^2 - 16x + 21 &= 0 \\ \Rightarrow 3x^2 - 9x - 7x + 21 &= 0 \\ \therefore x &= 3, \frac{7}{3}\end{aligned}$$

43. If the standard deviation of some temperature data in $^{\circ}\text{C}$ is 5. If the data were converted into $^{\circ}\text{F}$, the variance would be

- (1) 81 (2) 57 (3) 36 (4) 25

Sol. Answer (1)

$$\frac{C}{5} = \frac{F - 32}{9}$$

$$\Rightarrow F = \frac{9}{5}C + 32$$

So, t_0 convert $^{\circ}\text{C}$ into $^{\circ}\text{F}$ the each observation is multiplied by $\frac{9}{5}$ and 32 is added.

So, standard deviation will become $\frac{9}{5}$ times.

(as for addition no effect will cause)

$$\text{So, } \sigma = \frac{9}{5} \times 5 = 9$$

$$\text{So, } \sigma^2 = 81$$

44. If the standard deviation of a variable X is σ , then the standard deviation of variable $\frac{aX + b}{c}$ is

(1) $a\sigma$

(2) $\frac{a}{c}\sigma$

(3) $\frac{c}{a}\sigma$

(4) $\frac{a\sigma + b}{c}$

Sol. Answer (2)

If any variable is multiplied by any positive quantity then resulting standard deviation becomes that times and if there is some positive quantity added it brings no change to standard deviation.

45. If v is the variance and σ is the standard deviation then

(1) $v = \frac{1}{\sigma^2}$

(2) $v = \frac{1}{\sigma}$

(3) $v = \sigma^2$

(4) $v^2 = \sigma$

Sol. Answer (3)

46. If the S.D. of a set of observations is 8 and if each observation is divided by -2 , the S.D. of the new set of observations will be

(1) 4

(2) 8

(3) -4

(4) -8

Sol. Answer (3)

If each observation is divided by -2 then new S.D. will also get divided by same number

$$\text{i.e., } \sigma = \frac{8}{-2} = -4$$

47. Co-efficient of variation of two distributions are 50 and 60, and their arithmetic means are 30 and 25 respectively. Then, the difference of their standard deviation is

(1) 2.5

(2) 1.5

(3) 1

(4) 0

Sol. Answer (4)

$$\text{C.V.} = \frac{\sigma}{\bar{x}} \times 100$$

$$50 = \frac{\sigma_1}{30} \times 100$$

$$\sigma_1 = \frac{1500}{100} = 15$$

$$\text{So, } \sigma_1 - \sigma_2 = 0$$

$$60 = \frac{\sigma_2}{25} \times 100$$

$$\sigma_2 = \frac{1500}{100} = 15$$

48. The mean and variance of 6 observations are 5 and 7 respectively. If the four observations are 1, 3, 4, 6, then the remaining two observations are

- (1) 8 and 6 (2) 9 and 4 (3) 7 and 9 (4) 8 and 9

Sol. Answer (3)

$$\bar{x} = \frac{1+3+4+6+x+y}{6}$$

$$\Rightarrow x + y = 16$$

...(1)

$$\sigma^2 = \frac{(1+9+16+36+x^2+y^2)}{6} - (5)^2$$

$$\Rightarrow 7 = \frac{62+x^2+y^2-25 \times 6}{6}$$

$$\Rightarrow x^2 + y^2 = 42 + 150 - 62$$

$$\Rightarrow x^2 + (16-x)^2 = 130$$

$$\Rightarrow x^2 + 256 + x^2 - 32x = 130$$

$$\Rightarrow 2x^2 - 32x + 126 = 0$$

$$\Rightarrow x^2 - 16x + 63 = 0$$

$$\Rightarrow x^2 - 9x - 7x + 63 = 0$$

$$x = 7, 9, y = 9, 7$$

So, other two observations are 9 and 7.

49. Given that variance of 18 observations is 50. If each of the 18 observations is increased by 2, then variance of new data is

- (1) 50 (2) 52 (3) 48 (4) 25

Sol. Answer (1)

If we increases or decreases the term by any constant term then, there will be no effect on variance.

50. Given that variance of 30 observations is 20. If each of the observations is divided by 2, then new variance of the resulting observations is

- (1) 20 (2) 10 (3) 5 (4) None of these

Sol. Answer (3)

51. If $n = 10$, $\bar{x} = 12$ and $\sum x_i^2 = 1530$ then co-efficient of variation is

- (1) 36% (2) 41% (3) 25% (4) 38%

Sol. Answer (3)

$$\begin{aligned}\sigma^2 &= \frac{\sum x_i^2}{n} - \bar{x}^2 \\ &= \frac{1530}{10} - (12)^2 \\ &= 153 - 144\end{aligned}$$

$$\sigma^2 = 9$$

$$\begin{aligned}\text{So, C.V.} &= \frac{\sigma}{\bar{x}} \times 100 \\ &= \frac{3}{12} \times 100 \\ &= 25\%\end{aligned}$$

52. Co-efficient of variation of two distributions are 25 and 30 and their standard deviations are 480 and 630 then sum of their arithmetic means are

- (1) 3080 (2) 4020 (3) 4680 (4) 5650

Sol. Answer (2)

$$\begin{aligned}\therefore \text{C.V.} &= \frac{\sigma}{\bar{x}} \times 100 \\ \Rightarrow 25 &= \frac{480}{\bar{x}_1} \times 100 \\ \Rightarrow \bar{x}_1 &= 1920\end{aligned}$$

$$\begin{aligned}\text{and, } 30 &= \frac{630}{\bar{x}_2} \times 100 \Rightarrow \bar{x}_2 = 2100 \\ \text{So, } \bar{x}_1 + \bar{x}_2 &= 4020\end{aligned}$$

53. If the co-efficient of variation of a distribution is 70% and its standard deviation is 28, then its arithmetic mean is

- (1) 25 (2) 40 (3) 32 (4) 35

Sol. Answer (2)

$$\begin{aligned}\text{C.V.} &= \frac{\sigma}{\bar{x}} \times 100 \\ \Rightarrow 70 &= \frac{28}{\bar{x}} \times 100 \\ \Rightarrow \bar{x} &= \frac{28}{70} \times 100 \\ \therefore \bar{x} &= 40\end{aligned}$$

54. The mean and standard deviation of 50 observations are 50 and 10. Later on its was decided to omit an observation which was incorrect by recorded as 99. Then the standard deviation of the remaining 49 observations are

- (1) 7 (2) 7.21 (3) 6.5 (4) 8

Sol. Answer (2)

$$\bar{x} = \frac{\sum x_i}{50}$$

$$\Rightarrow \sum x_i = 2500$$

$$\text{Correct } \sum x_i = 2500 - 99 = 2401$$

$$\text{Correct } \bar{x} = \frac{2401}{49} = 49$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2$$

$$\Rightarrow 100 = \frac{\sum x_i^2}{50} - 2500$$

$$\Rightarrow \sum x_i^2 = 130000; \text{ correct } \sum x_i^2 = 130000 - 99^2 = 120199$$

$$\begin{aligned} \text{Correct } \sigma^2 &= \frac{\sum x_i^2}{n} - \bar{x}^2 \\ &= \frac{120199}{49} - (49)^2 \\ &= 2453 - 2401 \\ &= 52.04 \\ \sigma &= 7.21 \end{aligned}$$

55. If the co-efficient of variation of two distributions are 60% and 75% and their standard deviations are 45 and 40, then the ratio of their arithmetic means is

(1) 45 : 32

(2) 9 : 10

(3) 7 : 8

(4) 7 : 10

Sol. Answer (1)

$$C.V._1 = \frac{\sigma_1}{\bar{x}_1} \times 100$$

$$\bar{x}_1 = \frac{\sigma_1 \times 100}{C.V._1}$$

$$\bar{x}_1 = \frac{45 \times 100}{60} = 75$$

$$\frac{\bar{x}_1}{\bar{x}_2} = \frac{75 \times 3}{160} = \frac{45}{32}$$

$$C.V._2 = \frac{\sigma_2}{\bar{x}_2} \times 100$$

$$\bar{x}_2 = \frac{\sigma_2 \times 100}{C.V._2} = \frac{40 \times 100}{75} = \frac{160}{3}$$

56. If two distributions have equal means, then
- (1) The distribution with lesser S.D. is more consistent
 - (2) The distribution with lesser S.D. is less consistent
 - (3) Their C.V.'s will also be equal
 - (4) None of these

Sol. Answer (1)

57. Coefficients of variation of two distribution are 15 and 20 and their means are 20 and 10 respectively. If their standard deviations are σ_1 and σ_2 then
- (1) $3\sigma_1 = 2\sigma_2$
 - (2) $\sigma_1 < \sigma_2$
 - (3) $2\sigma_1 = 3\sigma_2$
 - (4) None of these

Sol. Answer (3)

58. Given $\sum_{i=1}^{20} a_i = 100$, $\sum_{i=1}^{20} a_i^2 = 600$, $\sum_{i=1}^{20} b_i = 140$, $\sum_{i=1}^{20} b_i^2 = 1000$, where a_i , b_i denotes length and weight of an observation. Then which is more varying?
- (1) Length
 - (2) Weight
 - (3) Equal C.V.
 - (4) None of these

Sol. Answer (1)

59. The mean of 50 observations were calculated as 4.04 by Mohan who took by mistake 10 instead of 8 for one observation. The correct mean is
- (1) 4.08
 - (2) 3.95
 - (3) 4.01
 - (4) 4

Sol. Answer (4)

SECTION - B

Linked Comprehension Type Questions

Comprehension-I

Median of a distribution is the value of the variable which divides it into two equal parts. In case of individual observations $x_1, x_2, \dots, x_n$, if the number of observations is odd, then median is the value of $\left(\frac{n+1}{2}\right)$ th observation when the observations have been arranged in ascending or descending order of magnitude. In case of even number of observations median is the A.M. of the values of $\left(\frac{n}{2}\right)$ th and $\left(\frac{n}{2} + 1\right)$ th observations, arranged in ascending or descending order of magnitude.

The mode of distribution is that value of the variable for which the frequency is maximum.

1. Median of the distribution 8, 5, 7, 9, 13, 11, 20, 23, 25, 28, 27 is
- (1) 11
 - (2) 13
 - (3) 12
 - (4) 20

Sol. Answer (2)

2. Median of the distribution 25, 20, 11, 3, 12, 18, 17, 9, 21, 22 is
- (1) 17
 - (2) 18
 - (3) 17.5
 - (4) 19

Sol. Answer (3)

3. The mode of following series is

value (x): 40 44 48 52 56 62 64 72 76

frequency (f): 10 12 14 20 15 19 18 8 4

(1) 76

(2) 64

(3) 62

(4) 52

Sol. Answer (4)

Comprehension-II

The standard deviation (σ) of variate x is the square root of the A.M. of the squares of all deviations of x from the A.M. observations.

If x_i/f_i ; $i = 1, 2, \dots, n$ is a frequency distribution then $\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2}$, $N = \sum_{i=1}^n f_i$ and variance is the square of standard deviation. Coefficient of dispersion is $\frac{\sigma}{\bar{x}}$ and coefficient of variation is $\frac{\sigma}{\bar{x}} \times 100$.

1. For a given distribution of marks mean is 35.16 and its standard deviation is 19.76 then coefficient of variation is

(1) 1.778

(2) 0.562

(3) 177.8

(4) 56.2

Sol. Answer (4)

2. The standard deviation for the set of numbers 1, 4, 5, 7, 8 is 2.45 then coefficient of dispersion is

(1) 4.9

(2) 2.04

(3) 0.49

(4) 2.45

Sol. Answer (3)

SECTION - C

Assertion-Reason Type Questions

1. STATEMENT-1 : The A.M. of series 1, 2, 4, 8, 16, 2^n is $\frac{2^{n+1} - 1}{n + 1}$.

and

STATEMENT-2 : Arithmetic mean (A.M.) of ungrouped data is $\frac{\sum x_i}{n}$ where $x_1, x_2, \dots, x_n$ are n numbers.

Sol. Answer (1)

2. STATEMENT-1 :
- | | | | | | |
|-------|---|---|----|---|---|
| x : | 5 | 6 | 7 | 8 | 9 |
| y : | 4 | 6 | 12 | — | 8 |

for given data mean of x ($\bar{x}$) is found to be 7.3. The missing frequency is 10.

and

STATEMENT-2 : In case of a frequency distribution, $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$.

Sol. Answer (1)

3. STATEMENT-1 : The mode of the items given as 0, 1, 6, 7, 2, 3, 7, 6, 6, 2, 6, 0, 5, 6, 0 is 6.

and

STATEMENT-2 : Mode is the observation with maximum frequency, whenever the other observations have for less frequencies.

Sol. Answer (1)

4. STATEMENT-1 : The mean of the series $x_1, x_2, \dots, x_n$ is $\bar{x}$. If x_2 is replaced by λ then new mean is independent of λ .

and

$$\text{STATEMENT-2 : } \bar{x} \text{ (mean)} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Sol. Answer (4)

5. STATEMENT-1 : If the mode of a data is 18 and the mean is 24, then median is 22.

and

$$\text{STATEMENT-2 : Mode} = 3 \text{ median} - 2 \text{ mean}$$

Sol. Answer (1)

6. STATEMENT-1 : The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2 then median of new set remains the same as that of the original.

and

STATEMENT-2 : If the variable of a series are arranged in ascending or descending order, then the value of the middle variable is defined as median.

Sol. Answer (1)

7. STATEMENT-1 : The mean deviation of the numbers 3, 4, 5, 6, 7 about mean is 1.2.

and

$$\text{STATEMENT-2 : Mean deviation about mean} = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$$

Sol. Answer (1)

8. STATEMENT-1 : If $a_1, a_2, a_3, \dots, a_n$ are positive real numbers, whose product is a fixed number c , then

the minimum value of $a_1 + a_2 + \dots + a_{n-1} + 2a_n$ is $n(2c)^{\frac{1}{n}}$.

and

$$\text{STATEMENT-2 : A.M.} \geq \text{G.M.}$$

Sol. Answer (1)

9. STATEMENT-1 : If $a \leq x_i \leq b$ where x_i denotes the value of x in the i^{th} case for $i = 1, 2, \dots, n$, then $(b - a)^2 \geq \text{Variance}(x)$.

and

$$\text{STATEMENT-2 : S.D.} \geq \text{range}(b - a)$$

Sol. Answer (3)

10. STATEMENT-1 : The S.D. of 5 scores 1 2 3 4 5 is $\sqrt{2}$.

and

$$\text{STATEMENT-2 : S.D.} = \sqrt{\text{variance}}$$

Sol. Answer (2)

11. STATEMENT-1 : Variance of first n natural number is $\frac{n^2 - 1}{12}$.

and

STATEMENT-2 : S.D. of first n natural number is $\sqrt{\frac{n^2 - 1}{12}}$.

Sol. Answer (1)

12. STATEMENT-1 : The variance of the variates 112, 116, 120, 125, 132 about their A.M. is 48.8.

and

STATEMENT-2 : $\sigma = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{\sum_{i=1}^n f_i}}$

Sol. Answer (1)

13. STATEMENT-1 : If the S.D. of a variate x is 6, then the S.D. of $ax + b$ is $|a| \cdot 6$.

and

STATEMENT-2 : The variance and S.D. is not independent of change of scale.

Sol. Answer (2)

14. STATEMENT-1 : The weighted mean of first n natural numbers whose weights are equal is given by $\left(\frac{n+1}{2}\right)$.

and

STATEMENT-2 : If $\omega_1, \omega_2, \omega_3, \dots, \omega_n$ be the weights assigned to the n values $x_1, x_2, \dots, x_n$ respectively of

a variable x , then weighted A.M. is equal to $\frac{\sum_{i=1}^n \omega_i x_i}{\sum_{i=1}^n \omega_i}$.

Sol. Answer (1)

15. STATEMENT-1 : If each of n numbers $x_i = i$ is replaced by $(i + 1) x_i$, then the new mean becomes

$$\frac{(n+1)(5n+4)}{6}$$

and

STATEMENT-2 : A.M. = $\frac{\sum_{i=1}^n x_i}{n}$.

Sol. Answer (4)

